

Smart Grid Communications Technology

Engineering Technology

May, 2026

Smart Grid Communications Technology (A13121)

Short Title:	Smart Grid Comms Technology	
Faculty	Science and Computing	
Department	Engineering Technology	
Cluster	Data Communications	
Author	CLAIRE.FITZPATRICK	
Credits	5	Level: Intermediate

Description of Module / Aims

The aim of this module is to direct the student towards a professional level of project research and implementation and the ability to achieve a level of expertise in telecommunications superimposed on power transmission lines. This module provides an insight into Smart Grid systems where the student as member of a team, will develop and build a transducer node that will communicate over a local ELV 50Hz closed grid.

Programmes

ELEC-0042	BEng (Hons) in Electrical Engineering (WD_ETRIC_B)	3 / 5 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (WD_ETRIC_BI)	3 / 6 / M
	BEng (Hons) in Information Engineering (International) (WD_EEELC_BI)	3 / 6 / M
ELEC-0042	BEng (Hons) (WD_ETRIC_B)	3 / 5 / M
	BSc (Hons) in the Internet of Things (International) (WD_KINTT_BI)	3 / 6 / M

Indicative Content

- Software control
- Micro-controller programming and interfacing methods
- Data and power mixing and filtering techniques
- Printed circuit board design and build
- Data transmission theory specific to calculation & analysis
- Simulation techniques and resources

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Achieve a level of expertise in RS422/485 nodal communications and implementation.
2. Implement a specific PCB design application for node fabrication and build.
3. Evaluate various filtering methods for data removal and superposition on a 50/60 Hz system.
4. Develop methods to dynamically adjust bandwidth to accommodate variances in data demand, (emulating Smart Grid).
5. Analyse the resultant data and compare transmission rates, BER, and attenuation with theoretical derivation.
6. Present project outcomes to a peer group within engineering.
7. Demonstrate the professional, ethical, and effective communications of a professional engineer.

Learning and Teaching Methods

- RS422/485 methods, (design theory and practical)
- Printed Circuit Board design, fabrication and population, (self-directed learning and staged assessment)

- Signal mixing and filtering, (theory and practical)
- Data transmission and retrieval QoS (theory and practical)
- Self-directed learning, tutorials, moodle and recommended software

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>In-Class Assessment</i>	40%	1,2,3,4
<i>Presentation</i>	10%	6,7
<i>Project</i>	50%	1,2,3,5,7

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Huntington, J. *Show Networks and Control Systems*. : Zircon Designs Press, 2013.
- Mitzner, K. *Complete PCB Design Using OrCAD Capture and PCB Editor*. : Elsevier Inc., 2009.

Supplementary Material

- Janert, P. *Data Analysis with Open Source Tools*. : O'Reilly, 2010.
- Self, D. *The Design of Active Crossovers*. : Elsevier Inc., 2011.